

Trainings ▾

General Information

with Mechanically Actuated Injector

Note : Warranty repairs **must** be performed at a Cummins® Authorized Repair Location to accurately calibrate, test, and repair the fuel systems on Cummins® engines.

When removing the injectors from the engine, compare the injector assembly part number with the appropriate Control Parts List (CPL) assembly part number listed in the CPL manual, Bulletin 3379133.

Note : The injector plunger and the barrel are machined to a very precise tolerance and are available as a match-fit assembly **only**. Do **not** mix the plungers and the barrels. **Always** replace the plunger and the barrel as a matched assembly if the injector is rebuilt.

Note : The outer base circle (OBC) valve and injector setting procedure can be used for all fixed time PT™ and hydraulic variable timing/STC engines, regardless of CPL or build date.

Note : All STC injector locknuts are made of powdered metal. These locknuts are porous by nature. When air pressure is applied to the STC oil delivery system, it is normal to see air bubbles form on the locknut. This is acceptable leakage. However, an audible air leak from a crack or hole in the locknut is unacceptable.

The injector barrel, plunger assembly, and injector cup are the parts most commonly replaced when rebuilding an injector.

The barrel, plunger assembly, and plunger to cup seal **must** pass an injector leakage test. See the Injector PT™ Rebuild Manual, Bulletin 3379071.

The injector stop screw was designed for standardization purposes. The stop screw is used to adjust the top-stop setting of the injector, and also acts as a housing for the tappet assembly. The revised injector stop screw can be used in all previous STC injector assemblies.

KT injector cups were released for the KTA marine injector in September, 1987 (Engine Serial Number first 33112795).

Due to injector plunger wear, a more durable D4 material replaced the 4140 material on the 9.5 mm [0.37 in] plunger in November, 1987 (Engine Serial Number first 33113265).

The following injectors have been fitted to K38 and K50 engines. Although this is **not** a complete list, it does cover the basic injector configurations which have been used.

The Standard K (non-top-stop) injector was the first injector fitted to the KV Series engine.

Reduced Drain Flow injectors were released as follows:

Engine	Engine Serial Number	Release Date
KTTA50-C	33111642	November 15, 1986
KT50-C	33111665	November 15, 1986
KTA50-C	33111645	November 15, 1986
KTA50-G	33111204	November 15, 1986
KTTA38-G	33111291	November 15, 1986

The injectors contained the same parts as previous injectors, except for the barrel and plunger assembly.

The injector drain flow is decreased by reducing the drain passage in the reduced drain flow barrel. The lower drain flow reduces the total flow required. The reduction in flow allows the use of the PT™ (Type G) pump in place of the PT™ (Type H) pump. It also allows the use of a 19 mm [0.75 in] PT™ (Type G) gear pump on K38 engines that previously used the 32 mm [1.25 in] PT™ (Type G) gear pump.

⚠ CAUTION ⚠

Reduced drain flow injectors must be used in complete sets due to the differences in total flow between the previous standard and reduced drain flow injectors. Failure to do so can result in engine damage.

Reduced drain flow injector assemblies were released for STC and non-STC injector configurations.

A full top-stop injector was released for K series engines in July, 1987 (Engine Serial Number first 33112560). This injector replaced the K hydraulic variable timing/STC injector. The full top-stop injector performs the same function as the previous K hydraulic variable timing/STC injectors, advancing the injection timing by trapping engine oil in the injector's hydraulic tappet. However, the revised injector has an additional top-stop feature to help reduce overhead wear. A top cap is used to stop the upward movement of the STC tappet. This unloads the injector train briefly, allowing oil to lubricate the link and socket joints.

Several of the components of the previous K hydraulic variable timing/STC injector were redesigned in order to convert to the full top-stop design, and some additional components were added.

The revised K full top-stop STC injector required a different oil transfer tube. The older tube was 7.93 mm [0.3125 in] longer, and will **not** fit into the oil feed locknut of the revised injector.

The K family STC injector was released for service and production on single stage uprate engines in February, 1988 (Engine Serial Number first 33113700).

These injectors are a similar style to the full top-stop STC injector. However, there are several differences.

- Injector Adapter - The adapter has two major features which make it different from the K STC injector adapter. First, the adapter was modified so that the same o-ring can be used at all three o-ring locations. Second, the adapter has locations for two orifice plugs, one on either side of the adapter.
- Injector Cup - It has a revised style cup, with a larger diameter tip that improves strength and reliability.
- Injector Cup Retainer - This was designed to accommodate the revised style cup and to improve ease of assembly.
- Injector Plunger - The plunger blank is longer than the previous K STC plunger blank. The length was added to the minor diameter areas to allow more fuel capacity.
- Injector Plunger Coupling - The injector plunger coupling is shorter than the previous K STC plunger coupling to compensate for the longer plunger blank. The overall length of the two different plunger and coupling assemblies after crimping is approximately the same. Previous injector plungers are **not** interchangeable with the new style.
- Internal Plunger Link - A revised internal link was fitted. It is shorter than the internal K STC plunger link in order to compensate for the longer injector plunger. The overall length of the revised link is 59.31 mm [2.335 in].
- Injector Plunger Return Spring - The injector utilizes a revised plunger return spring due to the higher travel requirements.

Press-Fit plungers and ceramic inner links were released for the standard K STC injector assemblies. The combination of these two features improves injector reliability and durability.

Press-Fit plunger assemblies have a stronger joint between the coupling and plunger.

A revised oil feed locknut and top-stop screw were released for K STC injectors. The revised oil feed locknut has the appropriate total travel settings stamped on it. The revised top-stop screw is 1.78 mm [0.070 in] longer than the previous top-stop screw.

In the STC injector, injection timing is controlled by the STC hydraulic tappet. The tappet has an inner piston (plunger) and an outer piston (sleeve).

Note : The plunger and sleeve are machined to very precise tolerances. The assembly is matched by a selection process at the factory (match-fit). Never exchange or combine the plungers and the sleeves.

These tappet components work together with the injector plunger to control injection timing. The tappet assembly can be used with any STC top-stop nut because they are **not** matched.

The tappet assembly and injector are **not** matched sets. The tappet assembly can be used in any STC injector. Cummins Inc. recommends STC tappet disassembly **only** if there is evidence that debris caused by an engine failure has moved beyond the full-flow filters. Any debris in the tappet will lower the performance of the tappet.

The method of adjusting the injection timing and injector rocker lever travel on engines with STC is the same as the method on engines without STC. The specification for injector travel and injection timing on engines with STC is different than engines without STC.

The Premium K STC injector assembly supersedes and obsoletes all Standard and Family K STC injector assemblies.

This injector includes the following features:

- Press-fit barrel and plunger assembly
- Longer barrel
- Revised oil feed locknut
- Revised top-stop screw
- Ceramic inner link
- Stronger cup.

The revised injector is more reliable and durable than the Standard and Family K STC injectors without the press-fit barrel and plunger assembly and ceramic link.

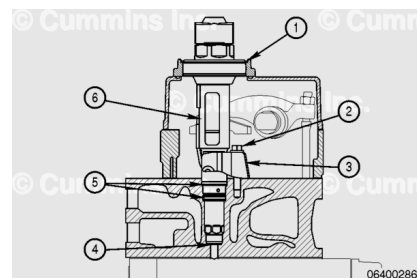
Standard K STC injectors can **not** be converted to a Premium K STC injector. Very few parts are interchangeable. The full injector assembly **must** be replaced.

Family K STC injectors can be converted to Premium K STC. However, no reliability or durability improvement will be gained if the Family K STC injectors have press-fit barrel and plunger assemblies and ceramic links. Press-fit barrel and plunger assemblies and ceramic links are available for Family K STC injectors.

with Electronically Actuated Injector

⚠ CAUTION ⚠

Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of these components can result in engine damage.



The injector consists of the following components:

1. O-ring/damper
2. Capscrew
3. Injector hold-down clamp
4. Seal
5. O-rings
6. Injector.

Initial Check

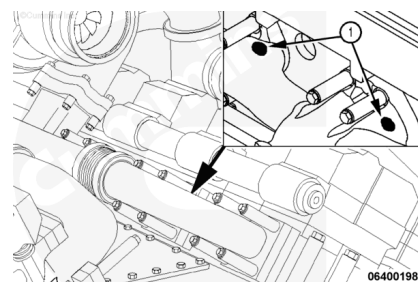
with Electronically Actuated Injector

Misfire Using Heat Indicator Marker

Note : This procedure will possibly **not** be effective when the symptoms occur only at low idle or when the engine is cold.

If the engine exhibits symptoms of a possible injector **not** firing, perform the following check:

- Operate the engine to normal operating coolant temperature of 80°C [180°F].
- Stop the engine and allow the exhaust manifold to cool for 10 minutes.
- Use a 204°C [400°F] temperature indicator marker, Part Number 3165163, to apply a mark (1) on the surface of all exhaust manifold ports. The mark **must** be as close to the cylinder head as possible, but **not** directly on the exhaust manifold mounting flange.
- Start the engine and operate under light load for a short time. Do **not** operate the engine under heavy load or for an extended period of time, as this can cause inaccurate results.

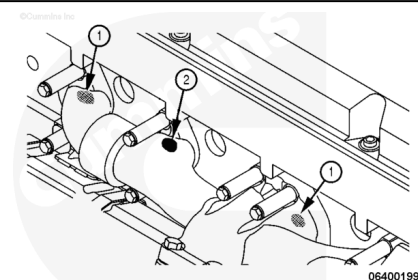


Note : It is **not** recommended to use the temperature indicator markers to troubleshoot for cylinders that are operating too hot.

If the 204°C [400°F] mark melts (1) the color will disappear and leave only a transparent glazed appearance where the mark was. This indicates that the cylinder is operating at a temperature above 204°C [400°F].

If the 204°C [400°F] mark does not melt (2) the mark color remains present. This indicates that the cylinder is not operating at or above a temperature of 204°C [400°F], indicating a possible injection malfunction.

Reference the Engine Performance Troubleshooting Tree for additional troubleshooting information.

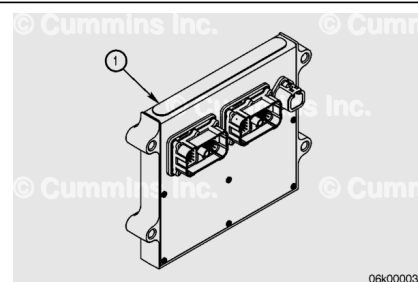


with Electronically Actuated Injector

Misfire Using Injector Testor Tool

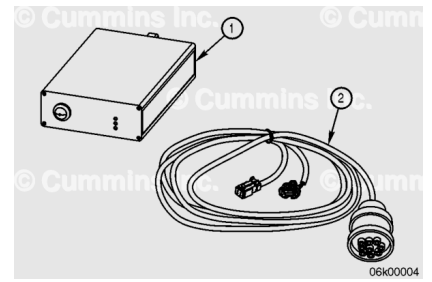
Note : This test **must** be performed with the engine shut down for at least 5 minutes.

Note : If the engine is equipped with CM2150 ECMs (1), cover the plug with a lint-free cloth. Slowly loosen the plug on the last injector of the high-pressure fuel lines 1/4 to 1/2 turn. The plug does **not** need to be removed to relieve the pressure. Connect INSITE™ electronic



service tool and verify the fuel pressure has bled down by monitoring the fuel rail pressure.

If the engine exhibits symptoms of an injector **not** firing, perform the following check using the Injector Tester (1)(2), Part Number 2892293:

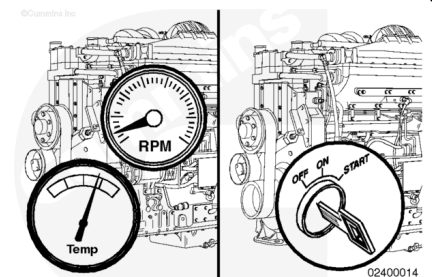


1. Plug the tool harness into the injector tester tool via the 4-pin connector.
 2. Plug the tool harness into the engine harness via the 9-pin connector.
 3. Verify the injector tester tool's yellow READY light is illuminated. If not, verify power supply to the 9-pin connector. If illuminated, proceed.
 4. Turn the keyswitch ON.
 5. Plug the tool harness into injector via the 2-pin connector.
 6. Fully depress and release the injector tester tool's PRESS TO TEST button.
- If the green PASS light is illuminated, the injector is functioning.
 - If the red FAIL light is illuminated, the injector is failed and must be replaced.

Repeat steps 5 and 6 until all injectors have been tested.

Misfire Using Infrared Thermometer

If the engine exhibits symptoms of an injector **not** firing, perform the following check:

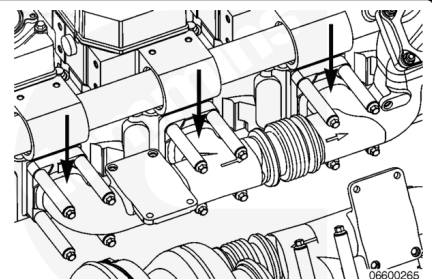


- Operate the engine to normal operating coolant temperature of 80°C [180°F].
- Once the engine has reached normal operating temperature, continue operating the engine and perform the following check.

Use an infrared thermometer, Part Number 3164487 or equivalent, to measure and record the surface temperature of the exhaust manifold ports.

Note : It is important to measure as close to the cylinder head as possible and in the same location on each of the exhaust manifold ports.

If an exhaust manifold port surface temperature is more than 40°C or 72°F (for electronically actuated injectors) or 93°C or 167°F (for mechanically actuated injectors) colder than the



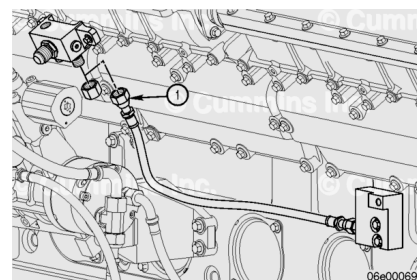
average surface temperature of all exhaust ports, this indicates a possible injector malfunction.

High-Pressure Injector Return Flow Test with Electronically Actuated Injector

Remove the injector drain flow hose from the fuel drain manifold block located on the left bank of the engine. Remove the hose from the drain manifold block end (1) **only**. Refer to Procedure 006-013 in Section 6.

(/qs3/pubsys2/xml/en/procedures/28/28-006-013.html)

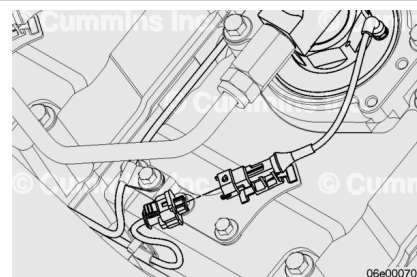
The union fitting in the fuel drain block **must** be capped to prevent fuel leakage during the test. It can be capped with Part Number 3164801, or equivalent.



Run the injector drain flow hose into a graduated cylinder to collect drain fuel flow. The cylinder must be capable of holding at least 300 ml [10 oz].



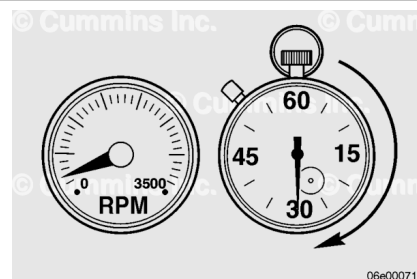
Disconnect each of the individual 2-pin injector electrical connectors (12 for QSK38 and 16 for QSK50). Use the following procedure in the QSK38, QSK50, and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-043 in Section 19 (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html).



With the return flow hose routed to the graduated cylinder, crank the engine at a minimum of 150 rpm for 30 seconds in 10 second intervals.

⚠ CAUTION ⚠

Do not overheat the starters. Damage to the starters can result.



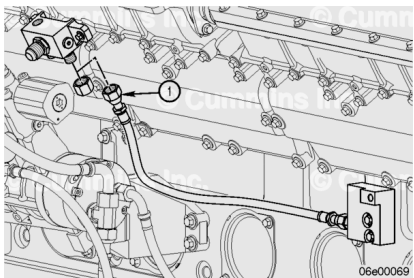
The volume of fuel collected during the 30 second crank cycle **must** be less than the volume indicated in the table below.

Measured Return Flow	
Engine	Volume
QSK38	150 ml [5 oz]
QSK50	200 ml [6.75 oz]

Remove the cap from the male union and install the fuel drain hose at the fuel drain block.

Tighten the hose.

Torque Value: 45 n•m [33 ft-lb]



High-Pressure Injector Return Flow Isolation Test with Electronically Actuated Injector

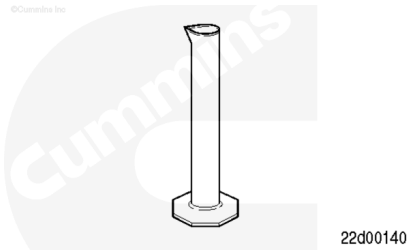
Drain the LTA cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)

Remove the intake manifolds. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html)

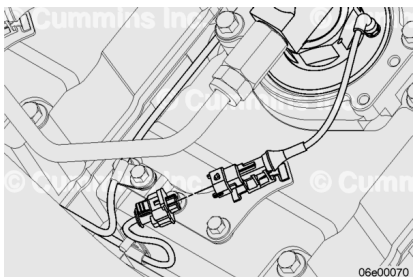


Install the fuel drain hose on cylinder number 1 LB in place of the banjo bolt through the fuel drain line that routes to the fuel drain block. The fuel drain hose for the QSK38 and QSK50 is Part Number 4918679.

Place the end of the fuel injector drain line fuel return flow hose in a small graduated cylinder to collect return flow.



Disconnect each of the individual 2-pin injector electrical connectors (12 for QSK38 and 16 for QSK50). Use the following procedure in the QSK38, QSK50, and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair

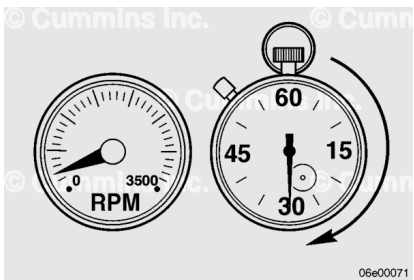


With the return flow hose routed to the graduated cylinder, crank the engine at a minimum of 150 rpm for 30 seconds in 10 second intervals.

⚠ CAUTION ⚠

Do not overheat the starters. Damage to the starters can result.

Record the amount of fuel collected.



Swap the fuel drain hose and banjo bolt between cylinders 1 LB and 2 LB.

Place the end of the fuel injector drain line fuel return flow hose in a small graduated cylinder to collect return flow.

Crank the engine at least 150 rpm for 30 seconds in 10 second intervals.

⚠ CAUTION ⚠

Do not overheat the starters. Damage to the starters can result.

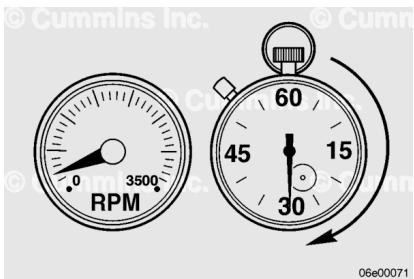
Record the amount of fuel collected.



Repeat the previous step until drain flow has been recorded for all cylinders on both banks.

Measurements		
	ml	fl-oz
Isolated Injector Return Flow with Engine Cranking at 150 rpm for 30 Seconds	15	0.5

⚠ CAUTION ⚠



Do not overheat the starters. Damage to the starters can result.

Replace injectors on any cylinder that has more leakage than specified above.

If any injectors were replaced, repeat the test on each of the cylinders until all cylinders have less flow than specified.

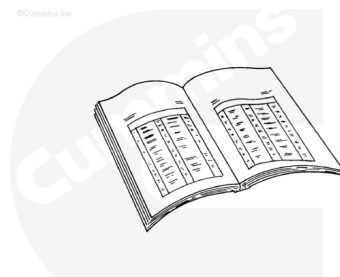


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Remove the fuel drain hose and assemble the fuel drain lines. Refer to Procedure 006-013 in Section 6 (</qs3/pubsys2/xml/en/procedures/28/28-006-013.html>).

Install the intake manifolds. Refer to Procedure 010-002 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-010-002-tr.html>)

Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)

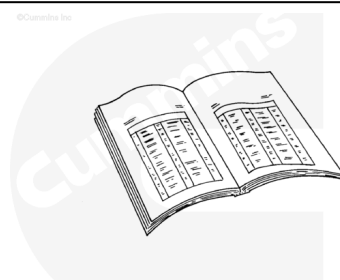


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Preparatory Steps

with Mechanically Actuated Injector

- Remove the rocker lever cover and all related parts. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html>)
- Remove the rocker lever assembly. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-009-tr.html>)



ck800wa

with Electronically Actuated Injector

- Remove the injector fuel supply lines. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html>)
- Install the injector protective plugs, Part Number 4918767.

- Remove the two-piece rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html>)

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Remove

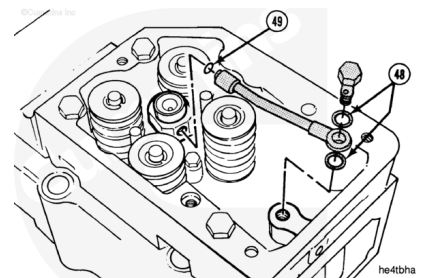
with Mechanically Actuated Injector

Note : This step applies to engines equipped with STC injectors.

Remove the STC banjo connector mounting screw.

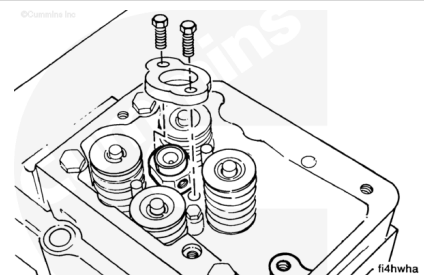
Remove the banjo connector.

Remove and discard the sealing washer (48) and o-ring (49).



Remove the injector hold down capscrew.

Remove the injector hold down clamp. Non STC **only**.



⚠ CAUTION ⚠

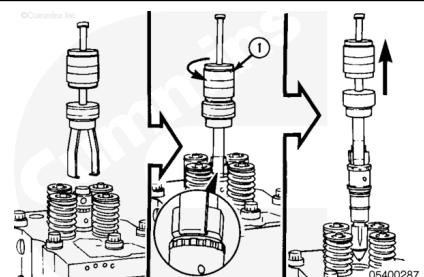
Do not allow the STC tappet to fall out of the injector. Engine damage can result.

Note : The rocker lever housing has been removed from the illustration for clarity.

For top-stop injectors, use injector puller, Part Number 3376497.

Make sure the puller arms are firmly under the top-stop screws.

Tighten the clamping ring (1).



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Use the slide hammer to remove the injector.

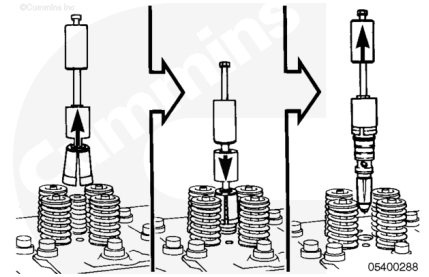
For standard injectors, use injector puller, Part Number 3376000 or 3376497.

Place the split collar over the injector.

Slide the locking collar over the injector

Use the slide hammer to remove the injector.

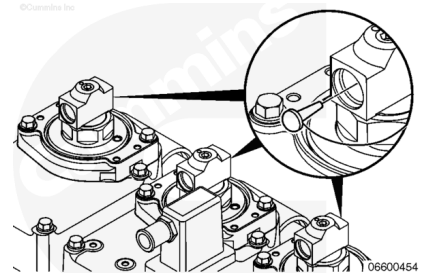
Note : If the injector pullers mentioned above are not available, carefully use a pry bar. Pry upward in the injector against the cylinder head.



with Electronically Actuated Injector

⚠ CAUTION ⚠

Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of these components can result in engine damage.

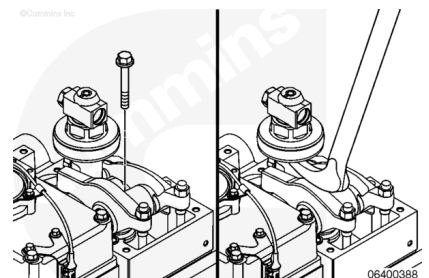


Note : Do not reuse protective plugs. The protective plugs must be used immediately upon removal from the plastic wrapping. If they are not used immediately, the protective plugs must be discarded. If the protective plugs are fouled in any manner before use, the protective plugs must be discarded and a new set used.

If **not** already completed, plug the ports in each side of the injector T-fitting with the injector protective plugs before removing the injector. Make sure the plugs are wedged firmly into place. This will prevent debris from entering and causing damage to the injector.

⚠ WARNING ⚠

To reduce the possibility of personal injury, place a clean rag around the injector as well as wear goggles or a face shield to prevent injury from fuel spray from residual fuel pressure while removing the injector.



⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.

Note : On applications with high drain head pressure, such as a fuel tank mounted higher than the engine, shut off the fuel drain line valve and evacuate the fuel drain line of fuel before removing the injector.

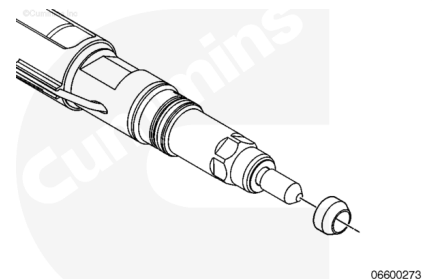
Remove and discard the injector hold-down capscrew.

Use a heel bar to remove the injector. Place the heel between the injector body and the exhaust rocker lever and gently pry the injector from the injector bore.

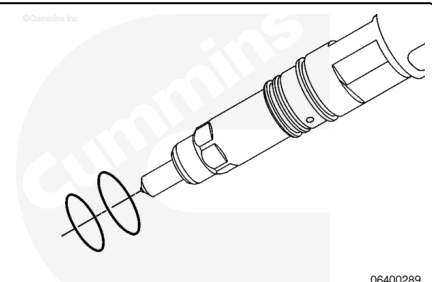
Note : On applications with high drain line head pressure, such as a fuel tank mounted higher than the engine, it is possible for fuel to leak from the drain line into the cylinder when the injector is removed. After removing the injector, check for fuel dripping into the cylinder and evacuate the fuel before continuing in the removal procedure.

Cover the injector bore in the cylinder head to prevent foreign objects from entering and possibly damaging the bore and combustion chamber.

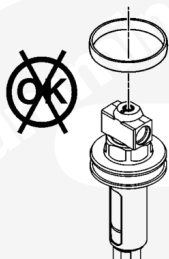
Remove and discard the seal on the end of the injector.



Remove and discard the two o-rings on the injector.



Do **not** remove the o-ring/damper at this time.



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Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

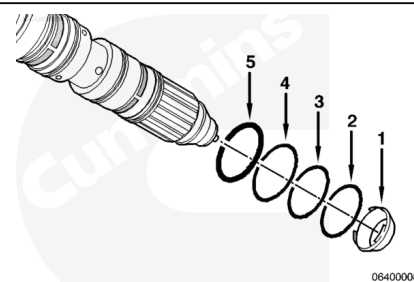
When using solvents, acid, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Injector plungers and barrels have a precise fit and are damaged easily. Only trained technicians are authorized to remove the plungers. Do not allow the plunger to fall out of the injector.



Note : Sealing rings are available in different thicknesses to adjust the injector protrusion.

Remove the o-rings (1), (2), and (3).

Remove the sealing ring (4) and record the thickness.

Clean the exterior of the injector with a lint free cloth.

Check the area that makes contact with the injector.

Check the orifice screen (5). It **must** be clean. If there is debris on the screen, remove the retaining clip and the screen for cleaning.

Clean the screen in solvent and dry with compressed air.

Install the screen and retaining clip.

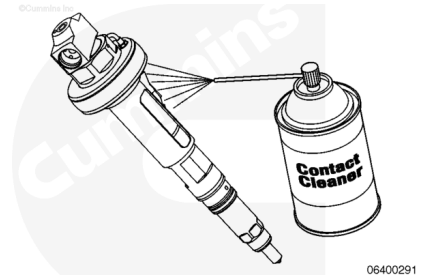
with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Clean the injector assembly with electrical contact cleaner, Part Number 3824510, or equivalent. Allow to air dry completely.

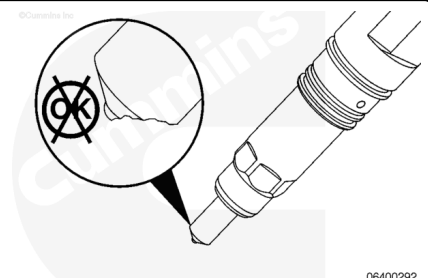
Do **not** use compressed air to dry or clean the injector.



Inspect the nozzle tip for damage.

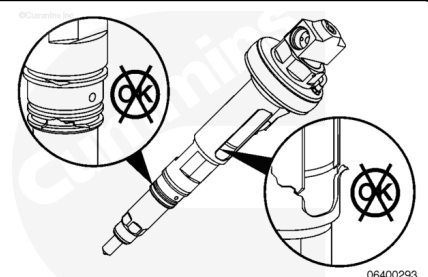
Replace injectors with damaged nozzles that prevent fuel delivery into the combustion chamber.

If the injector tip is damaged, inspect the cylinder bore and combustion chamber for damage as well.



Inspect the injector body for signs of corrosion, cracks, or any other damage.

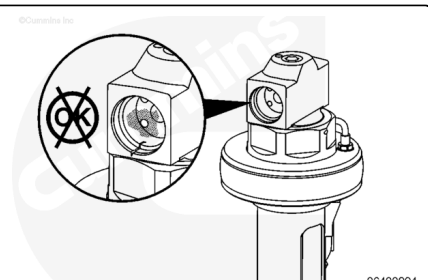
Inspect the o-ring grooves for damage that does **not** allow the o-rings to seal properly.



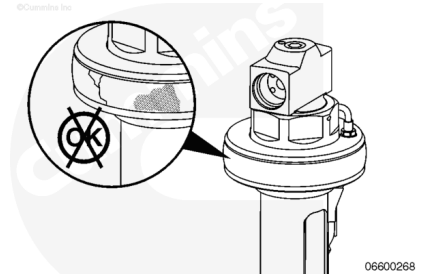
Inspect the T-fitting on top of the injector where the injector supply lines attach.

Inspect for corrosion, cracks and other damage that does **not** allow for proper sealing of the injector supply lines.

Replace damaged injectors.

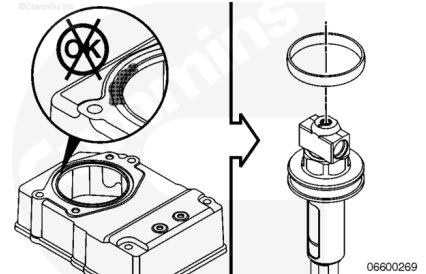


Inspect the o-ring/damper for wear, cracks, or other damage.
Replace, if damaged.



If the valve cover shows signs of oil leakage around the injector, replace the o-ring/damper.

Inspect the injector wiring for damaged wires, shrink tubing, elbow, and connector.

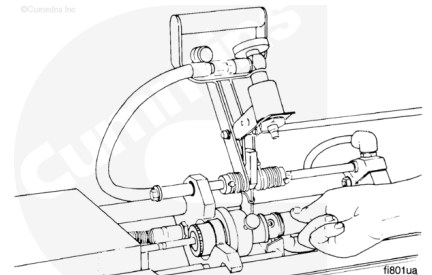


Calibrate

with Mechanically Actuated Injector

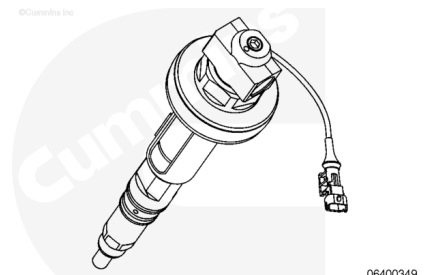
Note : The following information for the Mechanically Actuated Injector applies to the step timing control injector for the K38 and K50 engines.

This procedure requires special equipment and **must** be done at a Cummins® Authorized Repair Location.



with Electronically Actuated Injector

Calibration is **not** required for the Modular Common Rail Fuel System.

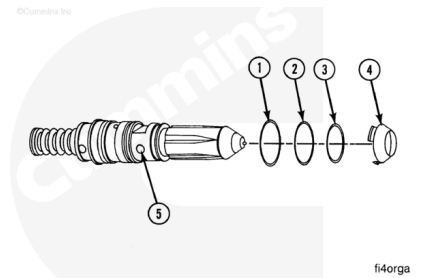


Install

with Mechanically Actuated Injector

Identify the o-rings so they can be installed in the correct groove.

- O-ring (1) has a red dot or stripe and is dull gray in appearance.
- O-ring (2) has no markings.
- O-ring (3) has a green dot or stripe. The o-ring has a shiny black appearance.

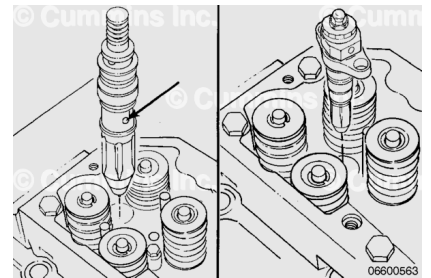


Lubricate the o-rings with vegetable oil and install them in the appropriate location.

Install the proper size seal ring (4).

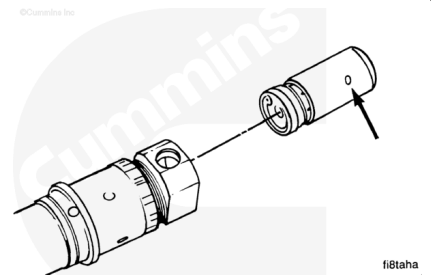
Position the standard injector in the bore. Turn the injector so that the injector screen points towards the hold down capscrew hole on the intake side of the cylinder head.

Note : Full top-stop injector and early top-stop injector plunger travel can **only** be adjusted when the injectors are removed from the engine. Use adjusting tool, Part Number 3822696, to set the injectors.



Do **not** allow the STC tappet to fall out of the STC (top-stop) injector. Damage can result.

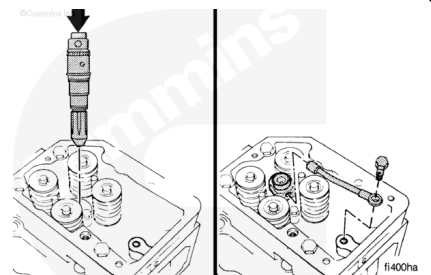
The STC tappet **must** be near the rocker lever assembly.



Do **not** push the injector on the seat until it is correctly aligned.

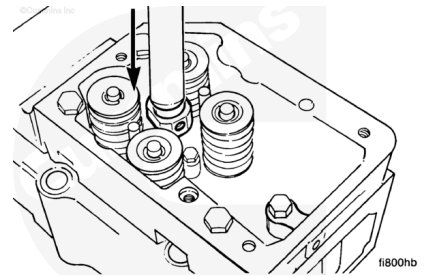
Place the STC injector in the bore. Turn the injector so the hole in the top-stop screw points to the oil supply hole in the rocker lever housing.

Use the oil jumper tube and the connector screw as tools. Turn the injector until the holes are aligned. Remove the connector screw and the tube.



⚠ CAUTION ⚠

Do not use a wooden tool to push the injector onto the seat. Failure can result because of splinters falling into the tappet.



Apply a quick hard push with a blunt object that touches the top-stop screw to seat the injector.

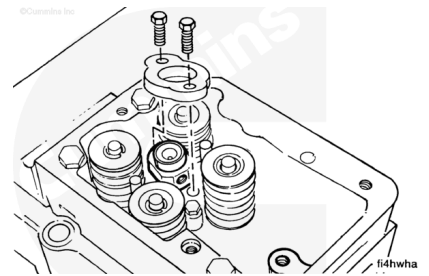
A single snapping sound will be heard when the injector is seated properly.

Install the hold-down clamp and the self-locking capscrews.

Tighten the capscrews alternately and evenly so the clamp will be centered on the injector body.

Torque Value: 16.2 n•m [145 in-lb]

Note : Engines equipped with STC injectors, have the hold down clamp already fitted to the injector.

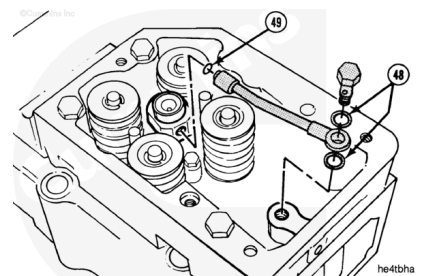


Install the o-ring (1) on the jumper tube. Lubricate the o-ring with clean engine oil. Install the jumper tube and the copper sealing washers (2).

Install the connector screw and tighten.

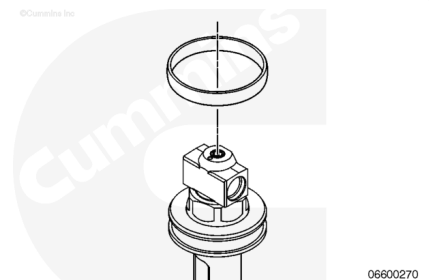
Torque Value: 25 n•m [221 in-lb]

Note : The oil jumper tubes for full stop STC injectors are 8 mm [5/16] shorter.

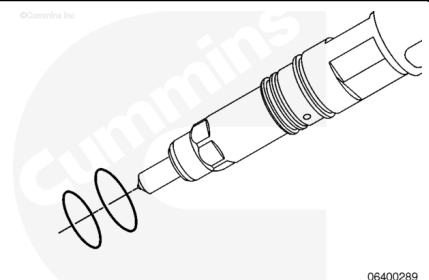


with Electronically Actuated Injector

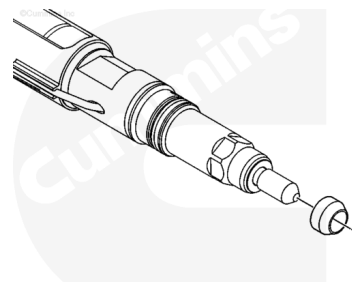
If the o-ring/damper is damaged, install a new o-ring/damper on the injector.



Install the new o-rings on the injector body.



Install the new seal on the end of the injector.



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⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

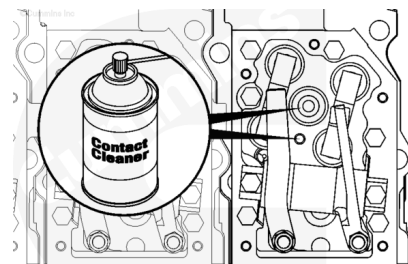
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of these components can result in engine damage.

⚠ CAUTION ⚠

Keep the injector bore and combustion chamber protected when using compressed air to prevent contamination. Failure to do so can cause engine damage.



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Note : On applications with high drain line head pressure, it is possible for fuel to leak from the drain line into the cylinder when the injector is removed. Before installing the injector, check for fuel dripping into the cylinder and evacuate the fuel before installing the injector.

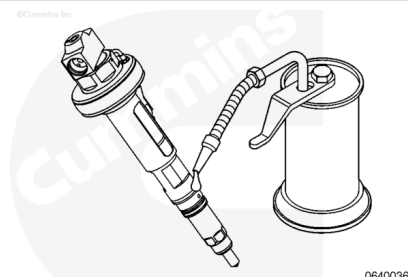
Clean the injector bore in the cylinder head with electrical contact cleaner, Part Number 3824510, or equivalent, for proper sealing of the o-rings.

Clean the threaded hole in the cylinder head where the injector hold-down capscrew will be installed with electrical contact cleaner, Part Number 3824510, or equivalent, and compressed air.

Lubricate the threads and underside of the new injector hold-down capscrew with clean lubricating engine oil.



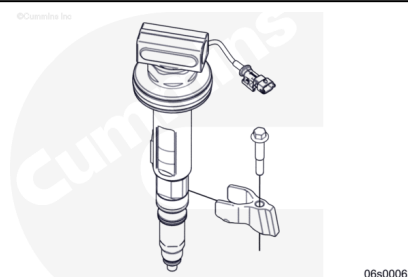
Lubricate all the o-rings with a light coating of clean lubricating engine oil.



⚠ CAUTION ⚠

Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of those components can result in engine damage.

Assemble the injector hold-down clamp and capscrew into the injector.

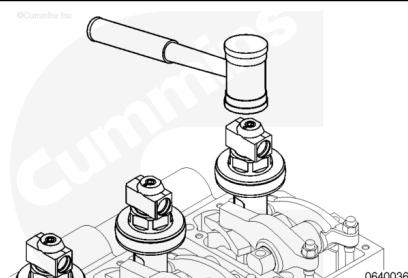


⚠ CAUTION ⚠

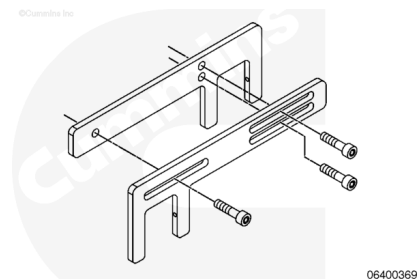
Only use a soft face mallet. Otherwise damage to the injector can occur.

The injector installer can be used on single or multiple injector installations.

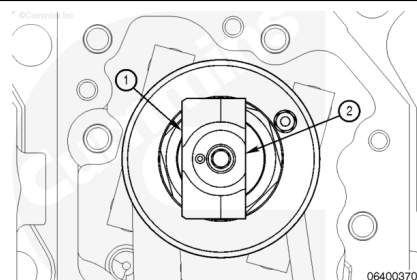
Install the assembly into the cylinder head by tapping gently on the injector with a soft face mallet.



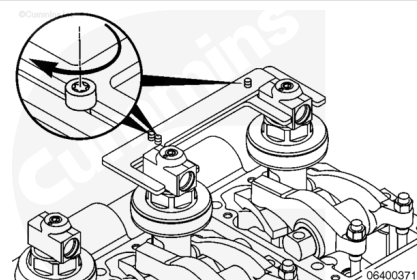
Assemble the injector installer kit, Part Number 4918620.
Finger tighten the three shoulder capscrews.



Place the injector installer on the injector such that the ball spring plunger on the installer is on the tapered side of the injector (1). If necessary, adjust the ball spring plunger so that it presses against the injector. Make sure the flat side of the injector (2) sits flush with the injector installer.



Align the injector installer with two injectors.
Tighten the three shoulder capscrews.

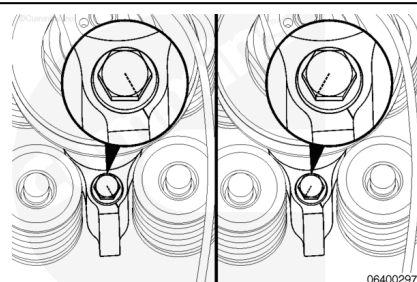


The 60 degree torque angle can be achieved by marking the location of one corner of the capscrew to a location on the injector hold-down clamp.

Torque Value:

Injector Hold-Down Clamp Capscrew

1. 68 n•m [50 ft-lb]
2. Loosen capscrew completely.
3. 30 n•m [22 ft-lb]
4. Rotate the capscrew 60 degrees.



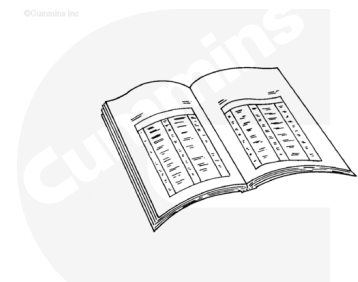
Finishing Steps

with Mechanically Actuated Injector

- Install the rocker lever assembly. Refer to Procedure 003-009 in Section 3.

(/qs3/pubsys2/xml/en/procedures/28/28-003-009-tr.html)

- Adjust the overhead. Refer to Procedure 003-006 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html)
- For STC Injectors, adjust the overhead. Refer to Procedure 003-007 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-007-tr.html)
- Install the rocker lever cover and all related parts. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)
- Operate the engine and check for leaks.



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with Electronically Actuated Injector

- Adjust the valves. Refer to Procedure 003-006 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html)
- Install the two-piece rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)
- Remove the injector protective plugs, Part Number 4918767.
- Install the injector fuel supply lines. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html)
- Operate the engine and check for leaks.